

# Backend Development Assignment

Project: The Task-Master API (RESTful Services)

## Project Overview

This assignment focuses on building a secure, scalable, and well-documented Task Management Backend. Candidates are expected to demonstrate proficiency in API design, database relationship management, and security implementation.

## Module 1: Server Initialization

### Environment Setup

*Set up the fundamental skeleton of the application.*

- Initialize project structure (Node/Express, Python/FastAPI, or Go).
- Implement `dotenv` for configuration management.
- Create a `GET /health` endpoint to verify server status.

## Module 2: Database Schema

### Relational Modeling

*Design a schema that handles user-specific data isolation.*

- **User:** Unique ID, Email (validated), Password (hashed).
- **Task:** Title, Description, Priority, Due Date, and Foreign Key to User.
- Implement database migrations or synchronization scripts.

## Module 3: API Specification

### Endpoint Development

*Build the CRUD interface for task management.*

| Method | Endpoint           | Action                                   |
|--------|--------------------|------------------------------------------|
| POST   | /api/auth/register | Create a new user account                |
| POST   | /api/tasks         | Create a task for the authenticated user |
| GET    | /api/tasks         | Fetch all tasks linked to the user       |
| PUT    | /api/tasks/:id     | Modify an existing task                  |
| DELETE | /api/tasks/:id     | Remove a task from the system            |

## Module 4: Security & Middleware

### Protection Layers

*Ensure the system is resilient against unauthorized access.*

- **Auth:** Implement JWT (JSON Web Token) Bearer authentication.
- **Hashing:** Use Argon2 or Bcrypt for password storage.
- **Validation:** Validate input payloads using schemas (e.g., Zod, Joi, or Pydantic).
- **Error Handling:** Implement a centralized exception filter to catch 404, 400, and 500 errors.

### Submission Requirements

- Source code hosted on a public/private Git repository.
- A `README.md` detailing installation and local execution steps.
- A Postman Collection or Swagger (OpenAPI) documentation.